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Course Name: SIGNALS AND SYSTEMS

Course Code: ECA 0304

EXPERIMENT – 9

OBJECTIVE

• To formulate a continuous-time LTI system from a differential equation, obtain its transfer function under zero initial conditions, simulate impulse and step responses, and relate the system poles to its transient behavior and stability.

Software / Laboratory Environment

MATLAB

Experimental Procedure

1. Write the governing first-order differential equation and identify its coefficients.
2. Apply the Laplace transform assuming zero initial conditions.
3. Form H(s) = 1/(s + 2) and enter numerator and denominator coefficients [1] and [1 2] in MATLAB.
4. Obtain the pole and zero of the system; verify that the pole is at s = −2.
5. Plot the impulse and step responses.
6. Apply a test input using lsim and observe the system response.
7. Relate the response shape and settling behavior to the pole at s = −2.

MATLAB Coding

```
clc; clear; close all;
% Continuous-Time LTI System: dy(t)/dt + 2y(t) = x(t)
% Assume zero initial conditions: H(s) = Y(s)/X(s) = 1/(s + 2)
num = [1];
den = [1 2];
sys = tf(num, den);
disp('Transfer function:');
sys
disp('Pole:');
disp(pole(sys));
```

```
disp('Zeros:');
disp(zero(sys));
t = 0:0.01:5;
figure;
impulse(sys, t);
grid on;
title('Impulse Response');
xlabel('Time (s)');
ylabel('h(t)');
figure;
step(sys, t);
grid on;
title('Step Response');
xlabel('Time (s)');
ylabel('y(t)');
% Response to x(t) = sin(2t)u(t)
x = sin(2*t);
figure;
lsim(sys, x, t);
grid on;
title('Response to sin(2t)u(t)');
xlabel('Time (s)');
ylabel('y(t)');
```

Input and Output Graphs

```
clc; clear; close all;
% Continuous-Time LTI System: dy(t)/dt + 2y(t) = x(t)
% Assume zero initial conditions: H(s) = Y(s)/X(s) = 1/(s + 2)
num = [1];
den = [1 2];
sys = tf(num, den);
disp('Transfer function:');
sys
disp('Pole:');
disp(pole(sys));

disp('Zeros:');
disp(zero(sys));
t = 0:0.01:5;
figure;
impulse(sys, t);
grid on;
title('Impulse Response');
xlabel('Time (s)');
ylabel('h(t)');
figure;
step(sys, t);
grid on;
title('Step Response');
xlabel('Time (s)');
ylabel('y(t)');
% Response to x(t) = sin(2t)u(t)
```



